

Cosmic Flashes: The Mystery of Fast Radio Bursts

How mysterious cosmic flashes connect stars, galaxies, and cosmology



AST201 Summer 2026

Who am I?



Name: Isabella Rivera

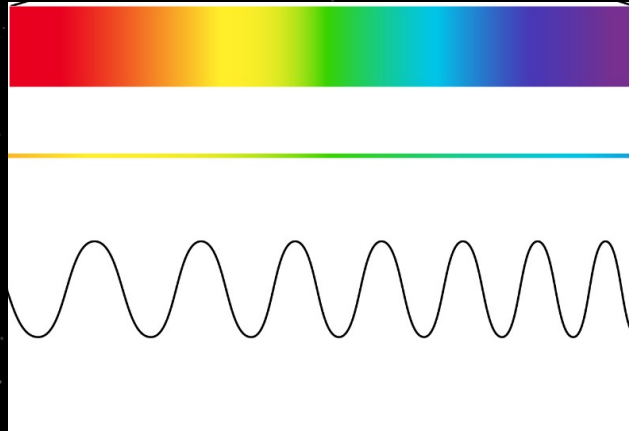
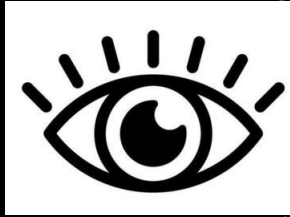
Program:

- ★ Specialist in Astronomy and Physics
- ★ Minor in Mathematics
- ★ Going into 4th year

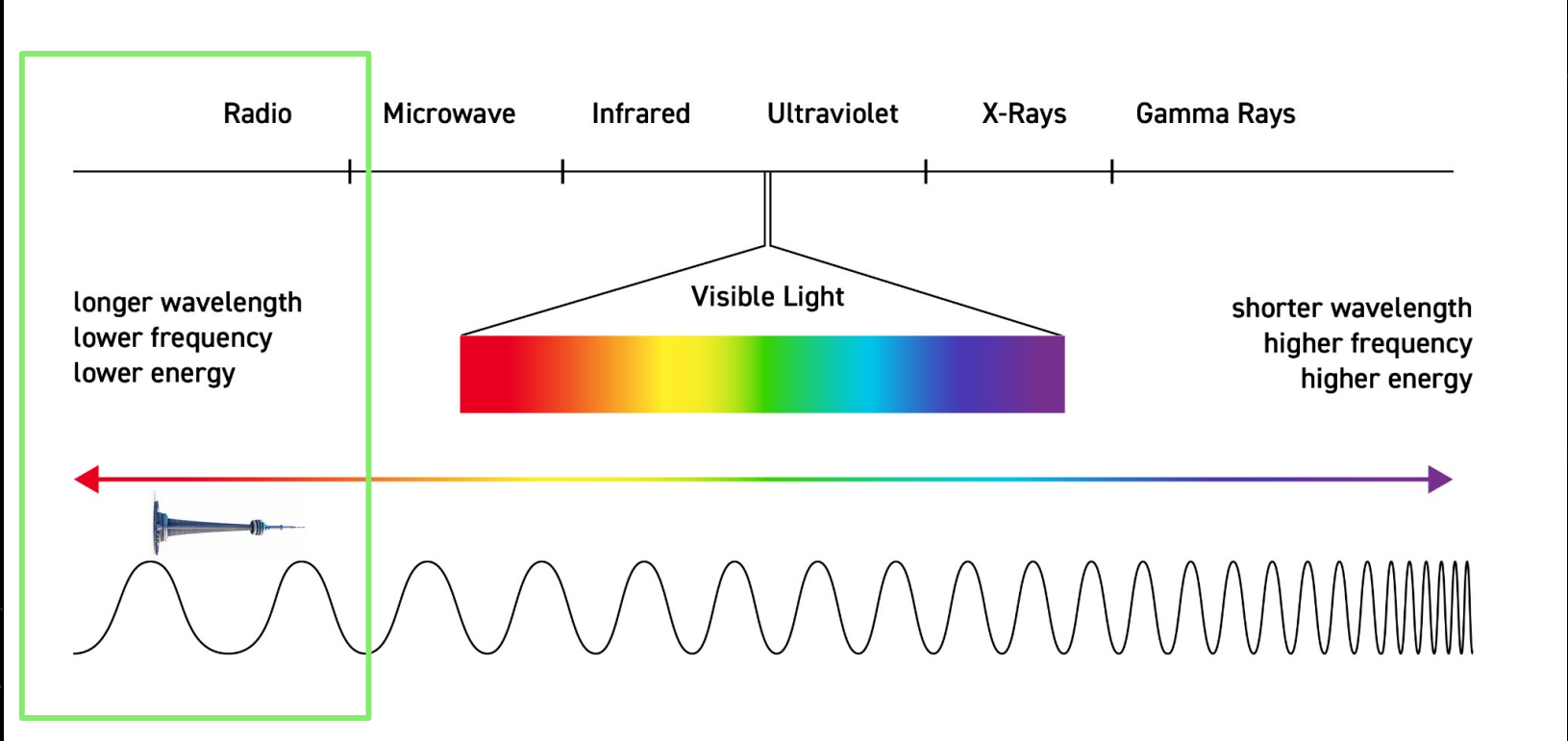
Research:

- ★ Fast Radio Bursts
- ★ Instrumentation
- ★ Stellar Evolution

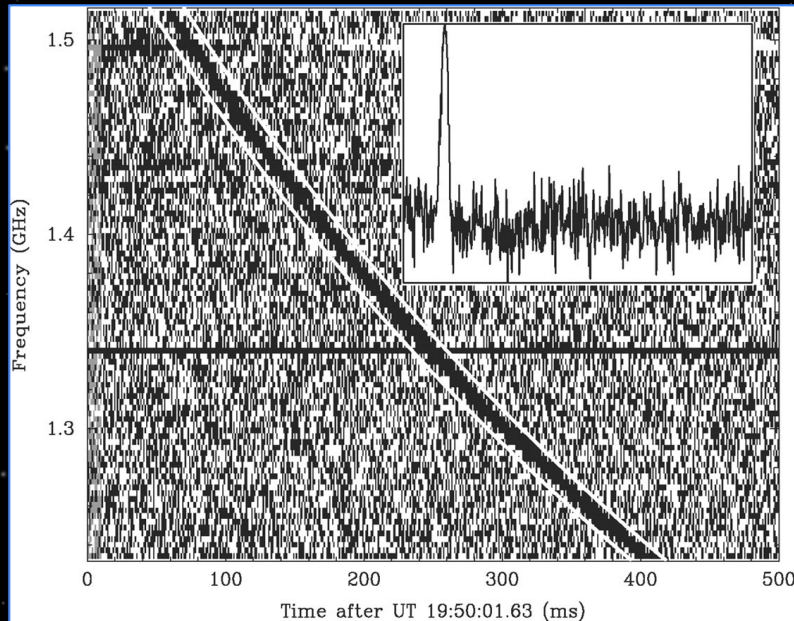
Electromagnetic Spectrum



Electromagnetic Spectrum



Fast Radio Burst (FRB)

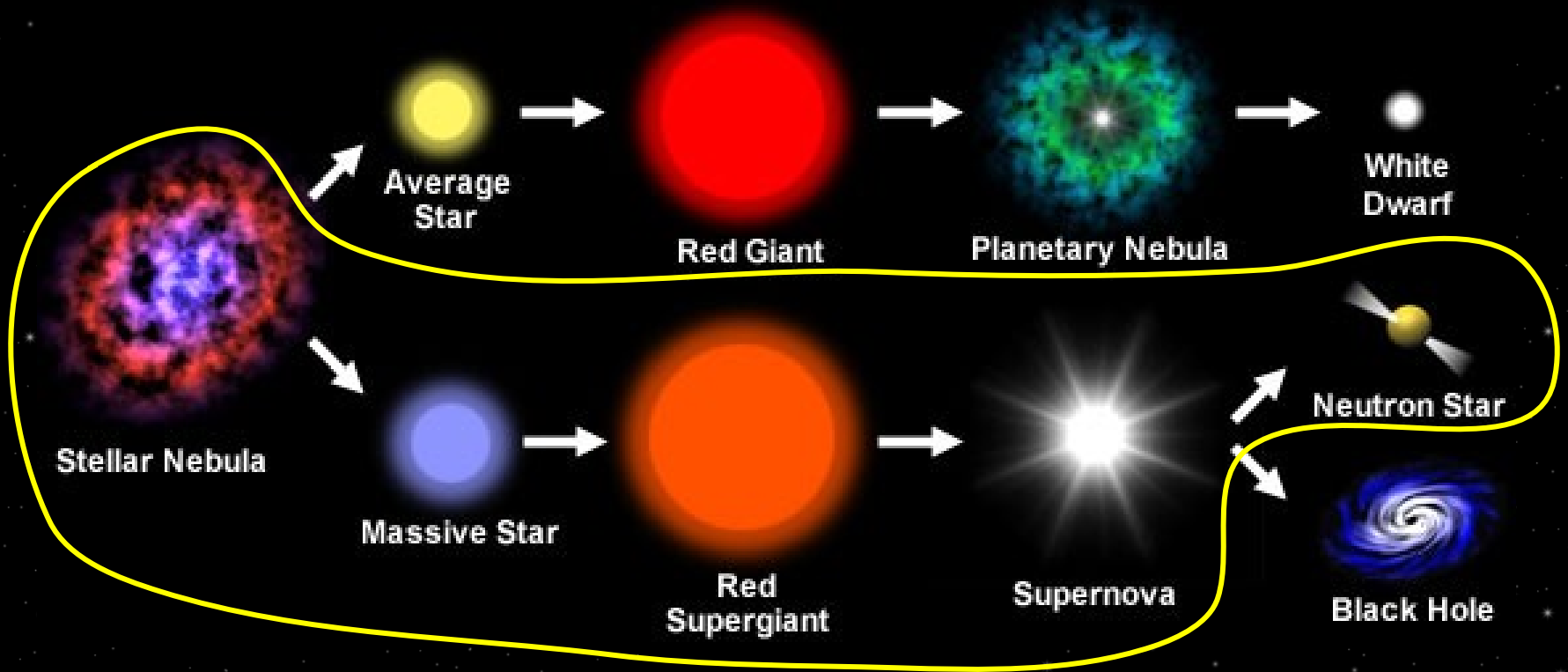


(Lorimer et al. 2007)

- ★ Astronomical radio pulses with **millisecond duration**
- ★ Discovered in 2007
- ★ More than 4,000 detected
- ★ **Extragalactic** (come outside our galaxy)
- ★ **Repetition?**

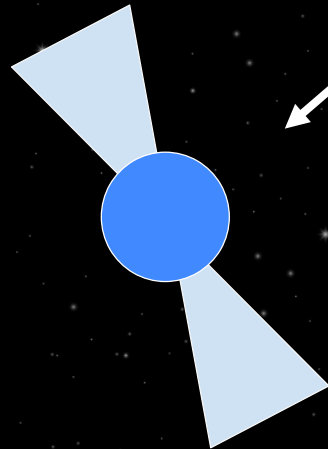
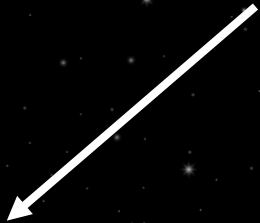
**What causes an
FRB?**

What do we know?

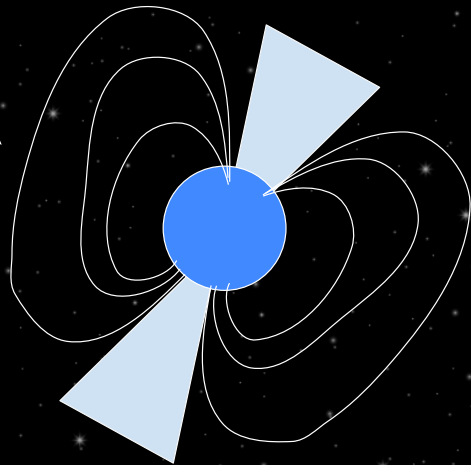




Neutron
Star

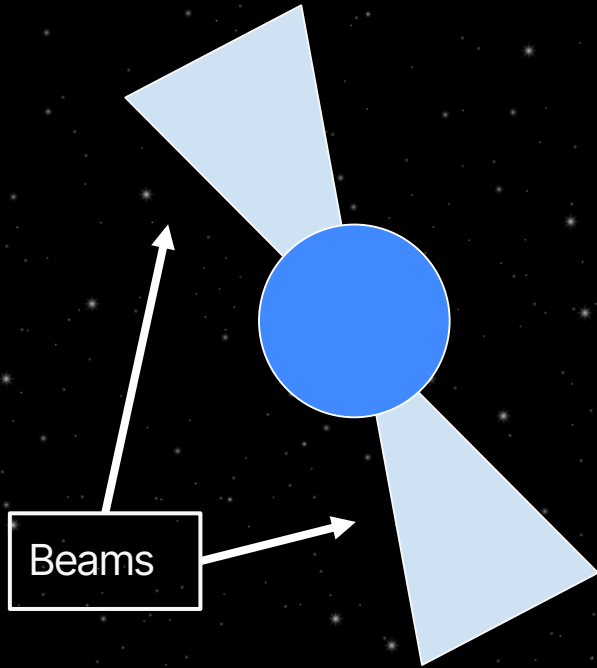


Pulsar



Magnetar

Type 1: Pulsar

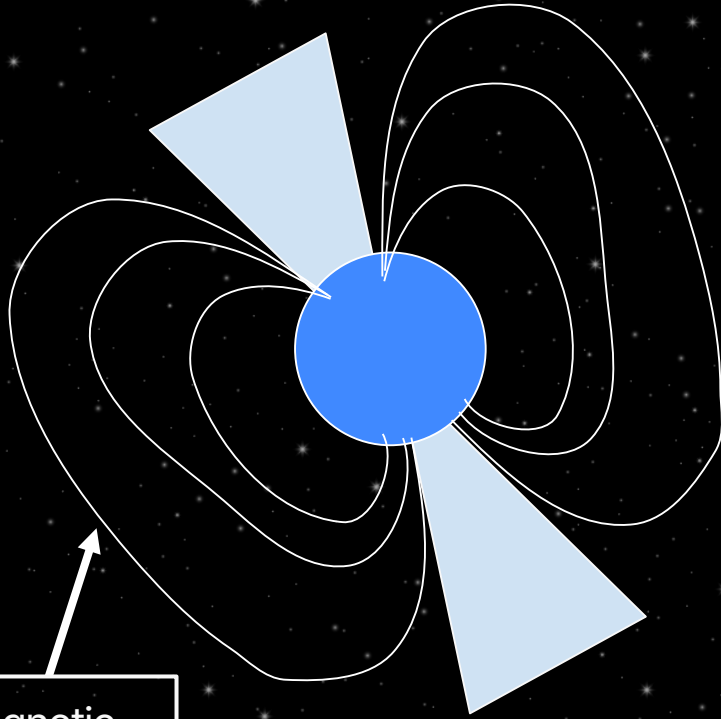


- ★ Neutron stars that are **highly magnetic and rotate rapidly**
- ★ Emit beams of radiation
- ★ We have discovered over 3000 pulsars
- ★ Spin rate ranges from hundreds of times per second to multiple hours
- ★ "**Lighthouses**" of the universe!

Type 1: Pulsar



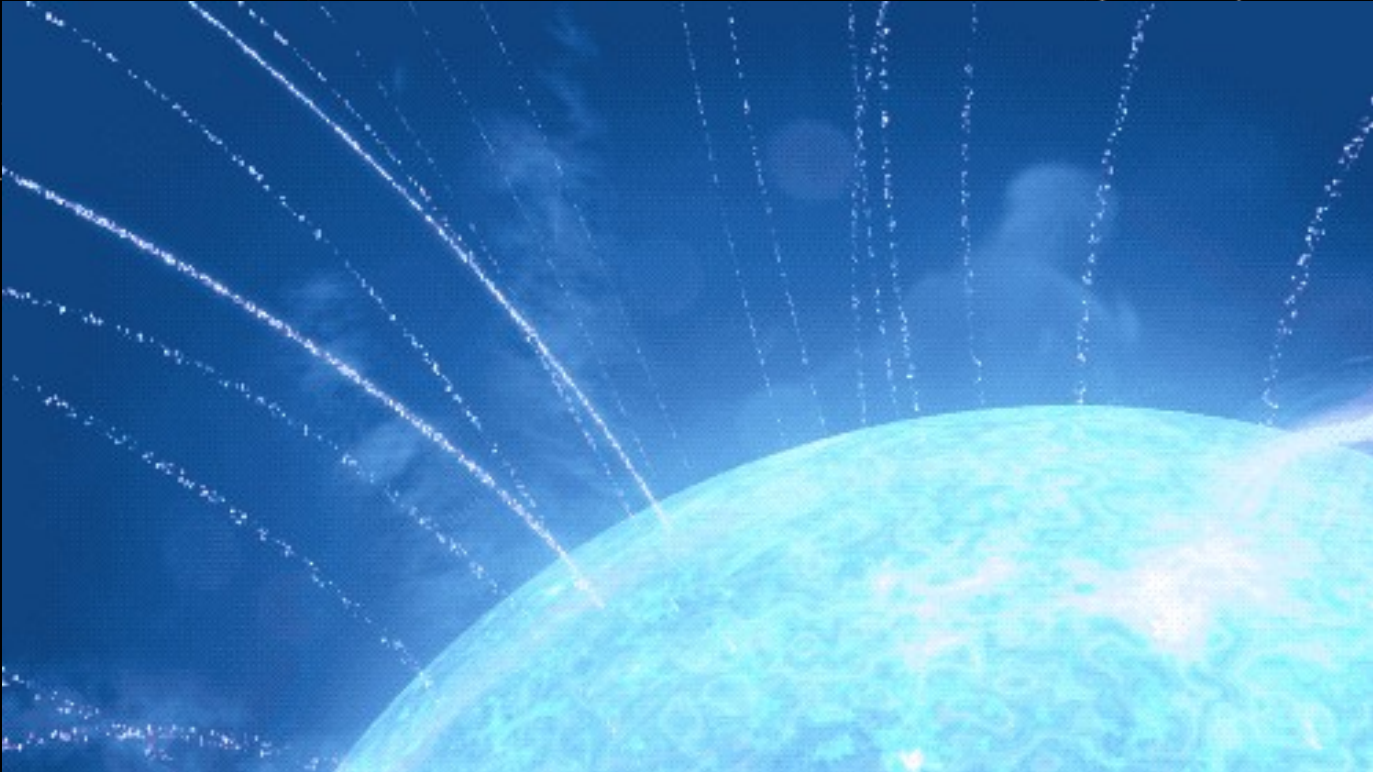
Type 2: Magnetar



Magnetic
Field Lines

- ★ Neutron stars that have a **magnetic field 1,000 times stronger** than that of a typical neutron star
- ★ Small movements in the crust result in the release of enormous amounts of energy
- ★ In just 0.1 seconds, magnetar SGR 1806-20 released more energy than the Sun emits in 100,000 years

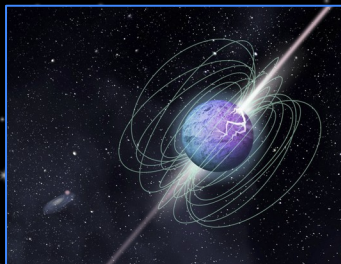
Type 2: Magnetar



What Produces Fast Radio Bursts?

Past Ideas

- ★ Neutron star mergers
- ★ Supernovae
- ★ Black holes
- ★ Cosmic Strings
- ★ Aliens
- ★ Magnetars



Current Ideas

- ★ In 2020, magnetar SGR 1935+2154 produced an FRB-like burst
- ★ This showed that magnetars can produce some FRBs
- ★ We still do not know whether they produce ALL FRBs

At one point, there were more theories for what produced FRBs than FRBs we had discovered!

POLL: What could create an FRB?

1. Colliding stars
2. Black holes
3. Aliens
4. Highly magnetized dead stars
5. We do not know yet

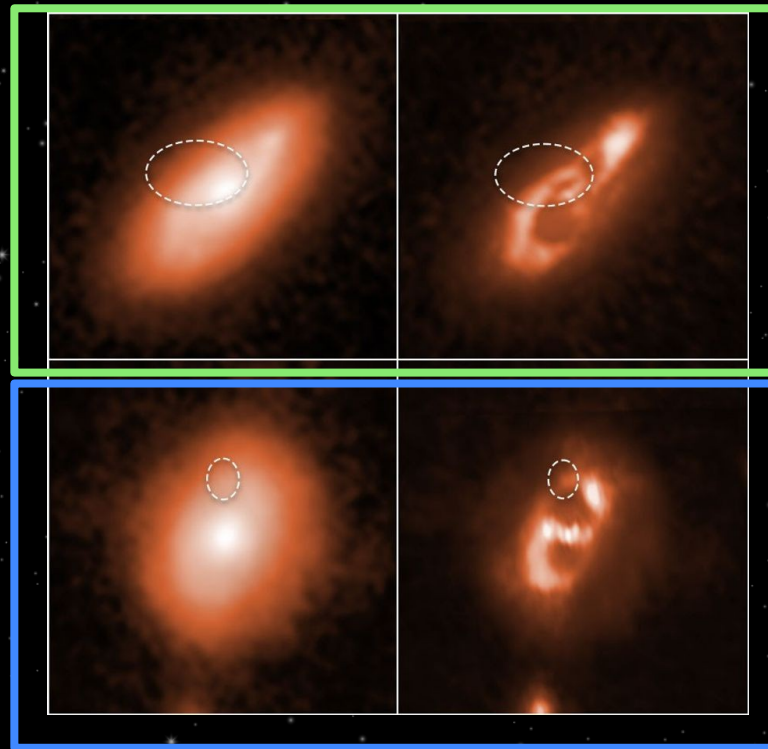
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**What kinds of
galaxies do we find
FRBs?**

FRBs come from a range of galaxies

- ★ FRBs can arrive from far beyond our Milky Way Galaxy
- ★ Astronomers need to find the burst's precise location on the sky
- ★ We can then identify its host galaxy
- ★ Host galaxies reveal the environments where FRBs are produced!



FRBs come from a range of galaxies



Spiral Galaxies

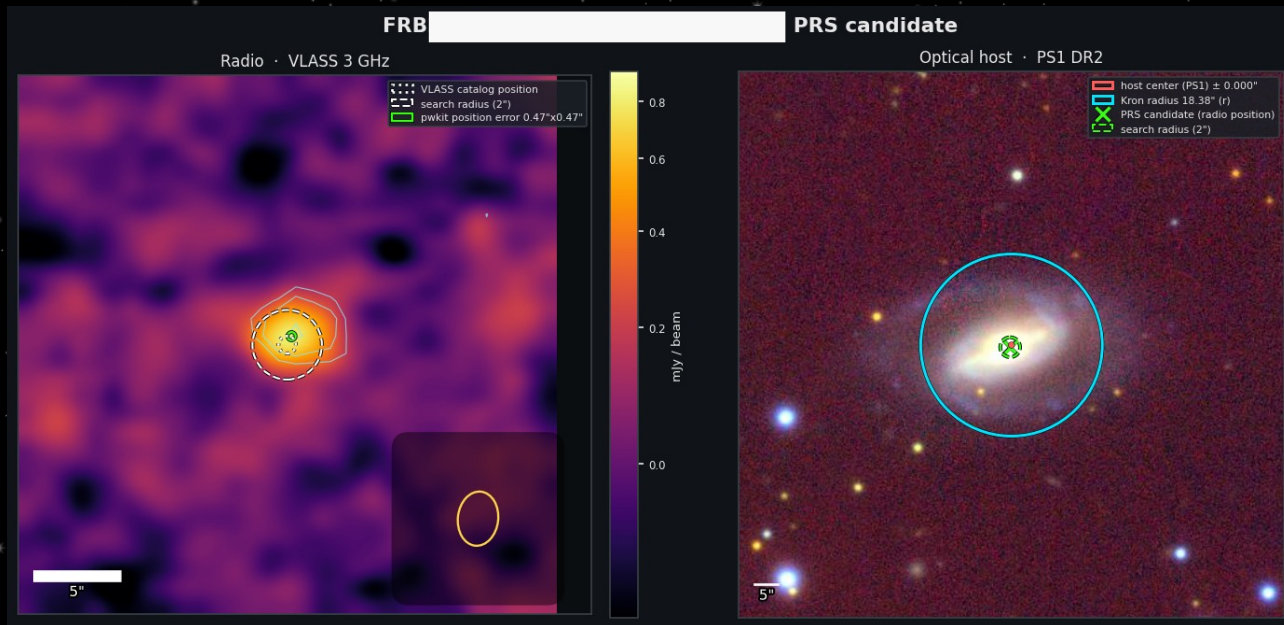


Elliptical Galaxies



Dwarf Galaxies

My research: Persistent Radio Sources (PRSs)



LEFT: A compact radio source seen in radio.

RIGHT: The radio source's position as seen in optical.

- ★ A steady radio source found at the same/near location as an FRB
- ★ We investigate whether the source is physically connected to the FRB

Why care about FRBs?

FRBs as Cosmic Probes

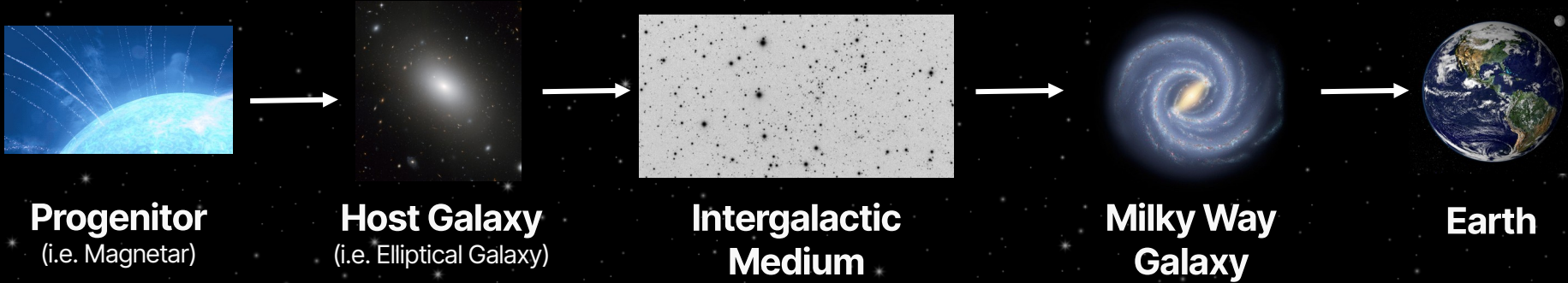


Progenitor
(i.e. Magnetar)



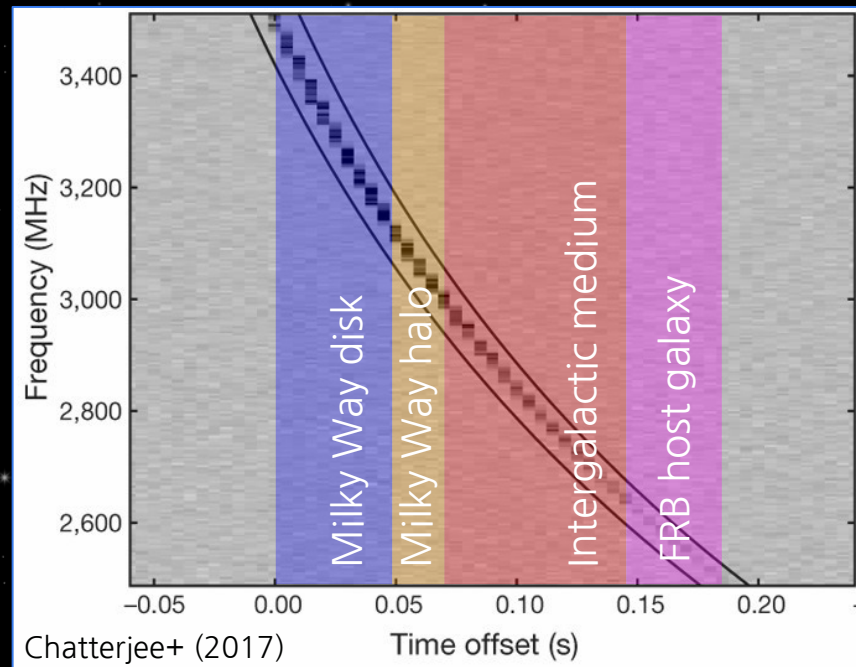
Earth

FRBs as Cosmic Probes



FRBs as Cosmic Probes

- ★ The space between galaxies is not completely empty
- ★ As the FRB travels, its signal gets delayed
- ★ This delay will teach us about how much material the burst passed through
- ★ FRBs can help study the matter between galaxies



(Credit to Dr. Ayush Pandhi)

How do we find FRBs?

CHIME



**Canadian Hydrogen Intensity Mapping Experiment
(CHIME)**

CHIME

Canadian Hydrogen Intensity Mapping Experiment (CHIME)

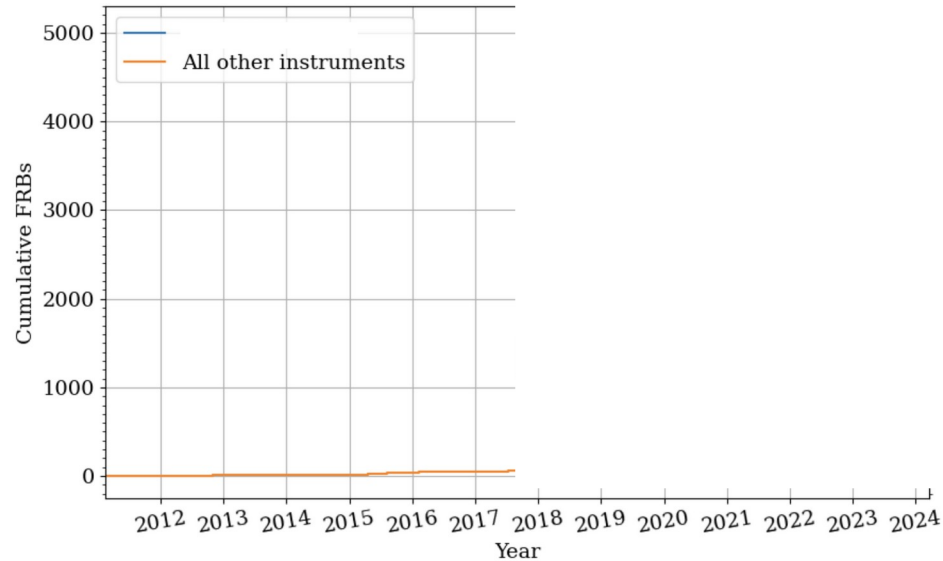


Image credit: CHIME Collaboration (via McGill Newsroom)

- ★ Located in Penticton, BC
- ★ Transit telescope with no moving parts
- ★ Four cylinders of 20 m x 100m in N-S orientation
- ★ Radio observing band
- ★ Great for FRB detection!

CHIME

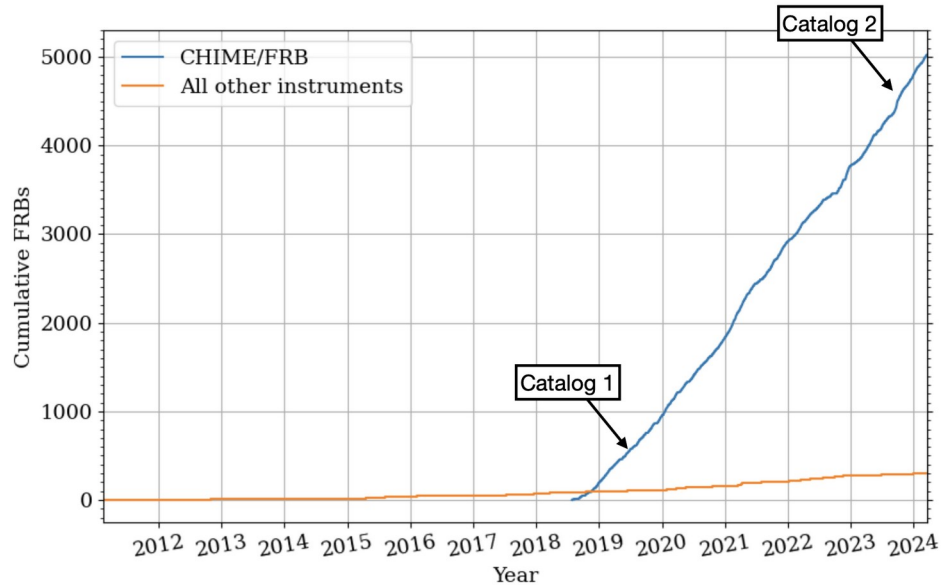
Number of FRBs Detected from 2012 - 2023 Globally



CHIME/FRB Collaboration 2025

CHIME

Number of FRBs Detected from 2012 - 2023 Globally



CHIME/FRB Collaboration 2025

POLL: What can an FRB teach us about?

1. Remains of massive stars
2. The galaxies that FRBs occur
3. Matter between galaxies
4. The large-scale Universe
5. All of the above

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ASX: Astronomy & Space Exploration Association

contact us at: space.association.utasx@gmail.com

Thanks!

Happy to take questions :)

Contact Information:
isabella.rivera@mail.utoronto.ca